

Theme: Startup Office IT Setup

❖ Section A: Scenario-Based Concept Application

1. Explain the selection process for CPU, RAM, and storage specifications to ensure hardware compatibility and prevent performance bottlenecks for a startup team performing concurrent development and video conferencing tasks.

ANS : For a startup where employees perform software development and video conferencing simultaneously, selecting compatible hardware is important.

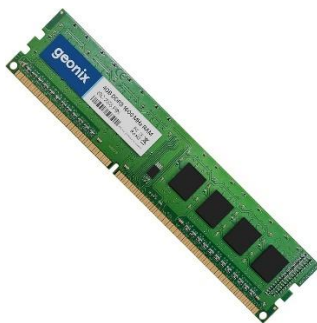
CPU

- Choose a multi-core processor (Intel Core i5/i7/i9 or AMD Ryzen 5/7).
- Minimum 6 cores and 12 threads.
- Supports virtualization and multitasking.



RAM

- Minimum 16 GB DDR4/DDR5.
- Prefer 32 GB for developers running virtual machines.
- Install identical RAM modules for dual-channel performance.



Storage

- Use NVMe SSD (512 GB or 1 TB).
- Faster than SATA SSD and HDD.
- Provides faster boot time and application loading.

**Hardware Compatibility**

- CPU socket must match the motherboard.
- RAM type (DDR4/DDR5) must be supported.
- SSD interface (M.2 NVMe or SATA) should match motherboard slots.
- Power Supply (PSU) should provide sufficient wattage.

Result

Proper hardware selection prevents CPU bottlenecks, memory shortages, and slow disk performance while supporting smooth multitasking.

2. Explain how system resource allocation and hardware bottlenecks contribute to performance degradation after the deployment of resource-intensive background applications like antivirus software.

ANS :

CPU Usage

- Real-time scanning increases processor load.
- Slower application response.

RAM Usage

- Antivirus occupies memory.
- Less RAM available for user applications.

Disk Usage

- Background scanning continuously reads files.
- SSD/HDD usage reaches high levels.

Performance Degradation

- Slow boot time
- Application lag
- High CPU utilization
- Increased disk access
- Reduced system responsiveness

3. Explain the systematic troubleshooting process for resolving peripheral hardware failures caused by driver incompatibility or operating system conflicts when a device functions inconsistently across different workstations.

ANS :

Step 1

Verify physical cable and power connections.

Step 2

Test the device on another workstation.

Step 3

Open Device Manager.

- Check for warning symbols.
- Identify missing or corrupted drivers.

Step 4

Install the latest manufacturer drivers.

Step 5

Update Windows/Linux.

Step 6

Restart the system.

Step 7

Check BIOS/UEFI if the hardware is disabled.

Step 8

Run hardware diagnostics.

Result

This process identifies whether the issue is caused by drivers, operating system conflicts, or faulty hardware.

4. Explain the role of Uninterruptible Power Supply (UPS) systems and surge protection in preventing file system corruption and ensuring data integrity during unexpected power fluctuations.

ANS :

➤ **UPS (Uninterruptible Power Supply)**

Functions:

- Provides backup power during outages.
- Prevents sudden shutdowns.
- Allows users to save work safely.

➤ **Surge Protector**

Functions:

- Protects hardware from voltage spikes.
- Prevents motherboard and PSU damage.

Benefits

- Prevents file corruption.
- Protects storage devices.
- Maintains data integrity.
- Increases hardware lifespan.
- Reduces unexpected system crashes.

❖ Section B: Practical Application

Task 1 : Physically assemble a workstation and document the component installation sequence.

ANS :

➤ **Component Installation Order**

1. Install CPU on motherboard.
2. Apply thermal paste.
3. Install CPU cooler.
4. Install RAM modules.
5. Install M.2 SSD (if available).
6. Install motherboard into cabinet.
7. Install Power Supply (PSU).
8. Connect 24-pin motherboard power cable.
9. Connect CPU 8-pin power cable.
10. Install HDD/SSD.
11. Connect SATA power and data cables.
12. Install Graphics Card (if required).
13. Connect front panel connectors.
14. Connect cabinet fans.
15. Perform cable management.
16. Power on system.
17. Verify successful POST.

Task 2 : Troubleshoot and resolve a RAM mismatch error within the BIOS/UEFI settings.
ANS :
➤ Steps

1. Restart PC.
2. Enter BIOS/UEFI (Delete/F2).
3. Verify RAM capacity.
4. Check RAM frequency.
5. Enable XMP/DOCP profile if supported.
6. Reseat RAM modules.
7. Install identical RAM modules.
8. Save BIOS settings.
9. Restart system.
10. Confirm correct RAM detection.

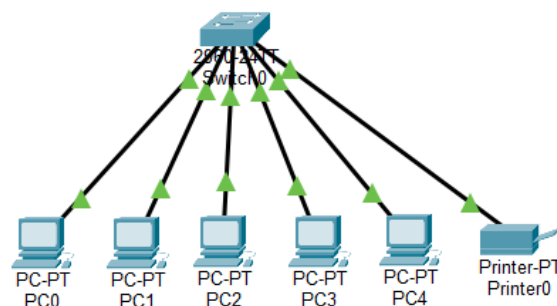
Task 3 : Build a LAN topology in Packet Tracer for 5 systems and 1 printer using a Class C subnet.
ANS : Devices

- 5 PCs
- 1 Printer
- 1 Switch

Network

 Subnet: **192.168.10.0/24**

Device	IP Address
PC1	192.168.10.1
PC2	192.168.10.2
PC3	192.168.10.3
PC4	192.168.10.4
PC5	192.168.10.5
Printer	192.168.10.10



Task 4 : Perform a clean OS installation with manual disk partitioning and startup optimization. CompTIA N+ [Security, Maintenance & Troubleshooting - A1]

ANS :

➤ **Steps**

1. Boot using Windows/Linux ISO.
2. Select Custom Installation.
3. Delete existing partitions.
4. Create new partitions manually.

Example:

System Partition: 150 GB

Data Partition: Remaining Space

5. Format partitions.
6. Install Operating System.
7. Install chipset drivers.
8. Install LAN drivers.
9. Install graphics drivers.
10. Install Windows/Linux updates.
11. Disable unnecessary startup applications.
12. Restart PC.

➤ **Startup Optimization**

- Disable unwanted startup programs.
- Update drivers.
- Enable automatic updates.
- Install antivirus.
- Verify Device Manager shows no errors.

❖ Section C: Integrated Case Study

Scenario: At Nexus Innovations, three rendering workstations pass morning POST checks but restart randomly during heavy afternoon workloads. The units are stored in poorly ventilated desk cabinets.

Questions:

1. Identify the three hardware components most likely to cause random reboots under load.

ANS :

- CPU (Overheating)
- Power Supply Unit (PSU)
- Graphics Card (GPU)

2. Explain why the systems pass POST in the morning but fail in the afternoon.

ANS :

Morning temperatures are lower.

During heavy rendering:

- CPU temperature rises.
- GPU temperature increases.
- PSU heats up.
- Poor airflow traps heat inside cabinets.

Once temperatures exceed safe operating limits, thermal protection forces automatic restarts.

3. How would you use a stress-test tool to confirm a thermal or power issue?

ANS :

➤ Use stress-testing software such as:

- Prime95 (CPU)
- OCCT (CPU/PSU)
- FurMark (GPU)

➤ Steps:

1. Monitor temperatures using hardware monitoring software.
2. Start CPU/GPU stress test.
3. Observe CPU, GPU, and PSU behavior.
4. If temperature exceeds safe limits or the PC restarts, overheating or PSU failure is confirmed.

4. Propose three specific steps to remediate the frequent system restarts.**ANS :****➤ Step 1**

Improve cabinet ventilation.

- Increase airflow.
- Remove workstation from enclosed cabinet.

➤ Step 2

Clean dust.

- Remove dust from fans.
- Replace thermal paste.
- Verify CPU cooler operation.

➤ Step 3

Test and replace faulty PSU if voltage is unstable.

5. Provide a brief technical justification for why a PSU upgrade might (or might not) be necessary.**ANS :**

A higher-capacity PSU may be necessary if:

- Existing PSU cannot supply enough power during heavy CPU/GPU workloads.
- Voltage becomes unstable under load.
- The PSU is old or inefficient.

A quality PSU with adequate wattage provides stable power, improves system reliability, and reduces unexpected restarts.